

The “Gaussian Splatting” Revolution



arXiv

<https://arxiv.org> › cs

NeRF: Representing Scenes as Neural Radiance Fields for ...

by B Mildenhall · 2020 · Cited by 16025 — We describe how to effectively optimize neural radiance fields to render photorealistic novel views of scenes with complicated geometry and ... [See more](#)

NeRF: multilayer perceptron + positional encoding.



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Mip-NeRF 360: Unbounded Anti-Aliased Neural Radiance ...

by JT Barron · 2021 · Cited by 2710 — We present an extension of mip-NeRF (a NeRF variant that addresses sampling and aliasing) that uses a non-linear scene parameterization, online distillation, [Se...](#)

Mip-NeRF: “distilled” and anti-aliased.



arXiv

<https://arxiv.org> › cs

3D Gaussian Splatting for Real-Time Radiance Field ...

by B Kerbl · 2023 · Cited by 6863 — We develop a fast visibility-aware rendering algorithm that supports anisotropic splatting and both accelerates training and allows realtime rendering.

Equivalent quality, 1000x faster rendering.

Splat Regression Models

3DGS: solving an inverse problem, by fitting a splat model, using particle based optimization.

Splat Regression Models: a particle-based toolkit for low dimensional modeling.